

Stephen MacKenzie

Greater Seattle Area
LinkedIn Profile
GitHub
YouTube Channel

Summary

Experienced Software Development Engineer and Team Lead with over 20 years at Microsoft, specializing in C++ development, testing, and ISO standards compliance. Proven expertise in software development, quality assurance, and team leadership. Currently developing embedded libraries for safety-critical applications with a focus on MISRA compliance. Passionate about contributing to advanced safety systems in the automotive and aerospace industries.

Languages

- C, C++, ARM Assembly
- Rust (embedded development)

Embedded Platforms

- ARM Cortex M series: STM32, Tiva C microcontrollers
- Raspberry Pi

Development Tools

- IAR Embedded Workbench, Keil uVision
- STM32CubeMX (code generator)
- STM32CubeIDE, uVision Runtime Manager
- Code Composer Studio, PinMux, Resource Explorer, Driverlib
- Visual Studio, VisualGDB, VSCode, Vim, NeoVim
- Rust embedded crates, ST-Link, OpenOCD
- CLI environment (gdb + stlink or openocd)
- STM32 HAL, CMSIS Programming, Legacy StdPeripheral
- MISRA Warnings enabled

Embedded Programming Skills

- Advanced algorithms: Fast Fourier Transform, Automata, Finite State Machines
- Electronic sampling: analog-to-digital, digital-to-analog, temperature, and voltage sampling
- Data integrity techniques: wear leveling, CRC
- Custom allocators using fixed array-based heap managers (e.g., Valvano)
- Standard C++ Library containers (e.g., `std::vector`) with fixed array-based custom allocators
- Data structures optimized for embedded environments (stacks, queues, circular buffers, binary trees, priority queues, heaps, graphs, open-address hash tables)
- Algorithms derived from "Introduction to Algorithms" by Cormen, Leiserson, Rivest, and Stein, Sedgewick's Algorithms, and "Foundations of Computer Science" by Aho and Ullman
- Automated embedded testing using CLI-based build, flash, and test sequences
- Integration with STM32CubeMX for STM32 and PinMux/Resource Explorer for Tiva C series
- CMSIS DSP Libraries for signal processing and testing

Unique Experience

In-depth understanding of MISRA-C and MISRA-C++ standards, developed from 20+ years on the Microsoft Visual C++ Team specializing in ISO C++ standards, CERT compliance, static/dynamic analysis, and security.

Embedded Systems Skills

- General Operating Systems knowledge, Computer Organization, Binary Structure, and Embedded Startup Code Sequences
- Timers and Clocks, generated programmatically or using tools like STM32CubeMX
- Interrupt Handling, Fault Monitoring, and Recovery
- Peripheral Communication: GPIO, UART, SSI, I2C, Timer, PWM, ADC, Ethernet, CAN
- DMA streaming, Memory Management, Flash Memory, EEPROM, Power Optimization
- Atomic Operations, Concurrency, and Parallelism (C++ Coroutines, Rust's `async/await`)
- Debugging Tools: Keil uVision, STM32CubeIDE, Code Composer Studio, VisualGDB, CLI (gdb + st-link or openocd)
- Semihosting, UART Printing, ITM Tracing

Professional Experience

Self-Employed *Performing Artist* Feb 2021 - Present Bellevue, Washington, United States

- Performed classical guitar concerts for charity.
- Studied under Dr. Jon Yerby and achieved high-level performances.

Dell EMC *Executive Consulting Architect* Aug 2018 - Jan 2019 Greater Seattle Area

- Troubleshoot critical issues for Dell's acquisition of Isilon Systems.
- Worked with C/C++, Assembly, Unix, Linux, MySQL for high-level troubleshooting.

Director of a Non-Profit and Public Speaker *Embedded Systems, Public Speaking, C/C++, Assembly* May 2014 - Aug 2018 Various Locations

- Developed MLibs, an open-source suite of C and C++ libraries for embedded systems.
- Delivered talks on embedded systems and containerization.
- Conducted charity work in Phoenix, Sacramento, Portland, and Tacoma.

Microsoft *Software Engineer and Manager* 1993 - May 2014 Redmond, Washington, USA

- Led the development and testing of C++ Libraries and backend tools.
- Directed C++ security initiatives for Visual C++.
- Contributed to ISO-standard-compliant C++ library development.

Education

University of Washington Certification in Embedded and Real-Time Systems 2016 - 2017

Cornish College of the Arts Classical Guitar Studies

Skills

- Programming Languages: C++, C, Assembly, ARM Assembly, Python, Rust
- Embedded Systems: ARM, Cortex-M, IAR Embedded Workbench, GNU ARM Eclipse
- Development Tools: Visual Studio, Clang++, GCC, Make, CMake
- Software Development: Embedded Systems, Real-Time Systems, Safety-Critical Applications
- Standards Compliance: MISRA, ISO C++ Standards

Honors & Awards

- **Black Belt Award** - Microsoft Corporation, Jan 1995
- **Extra Mile Award** - Microsoft Corporation, Mar 1994
- **Washington State Programming Contest Winner**, Mar 1992

Projects

Open Source Contributions - MLibs RTOS Framework: Developed and maintained an open-source suite of C and C++ libraries for embedded systems.

Classical Guitar Performances - Performed benefit concerts, raising funds for charitable causes.